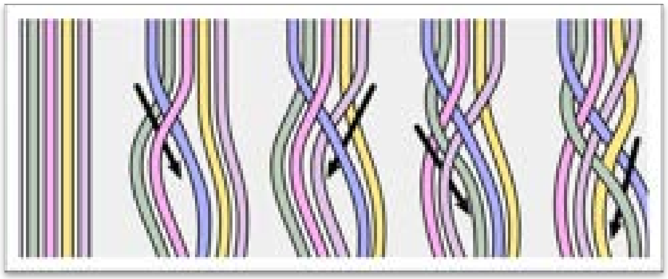
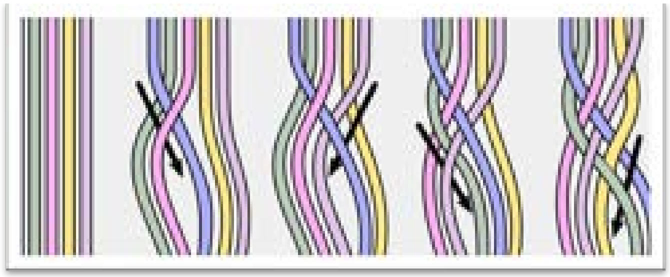
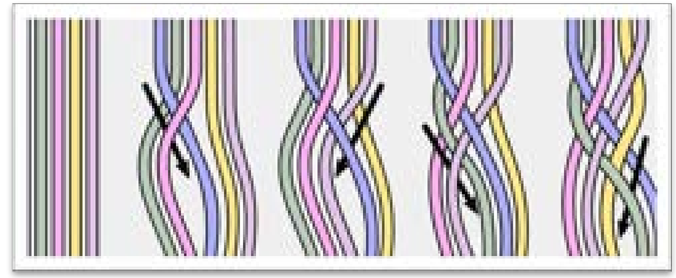
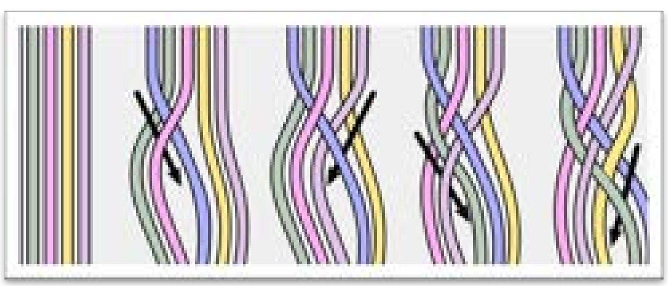
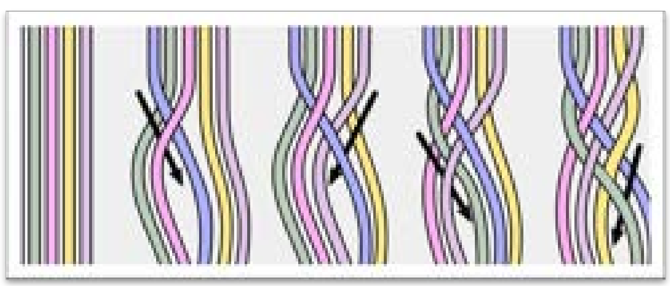
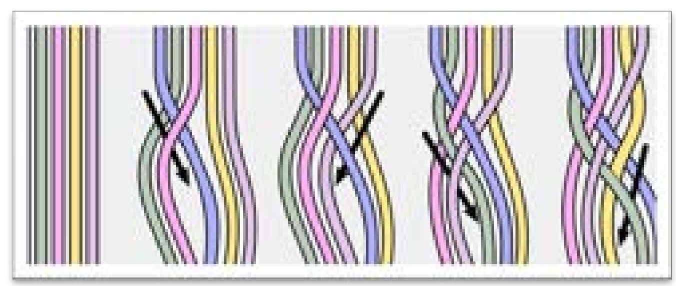




Braid Groups

Configuration Space Summer School 2026

Gahl Shemy

Recall from this morning...

Key players: (un)ordered configuration spaces!

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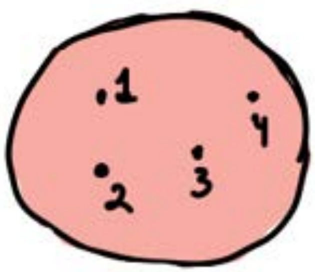
Let X be a top. space. The (ordered) configuration space of n points on X is given by

$$\mathrm{Conf}_n(X) := \{(x_1,\dots,x_n) \in X^n \mid x_i \neq x_j \ \forall i \neq j\}.$$

X^n without the "fat diagonal"



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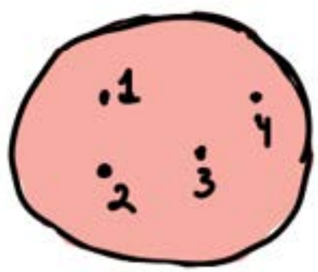
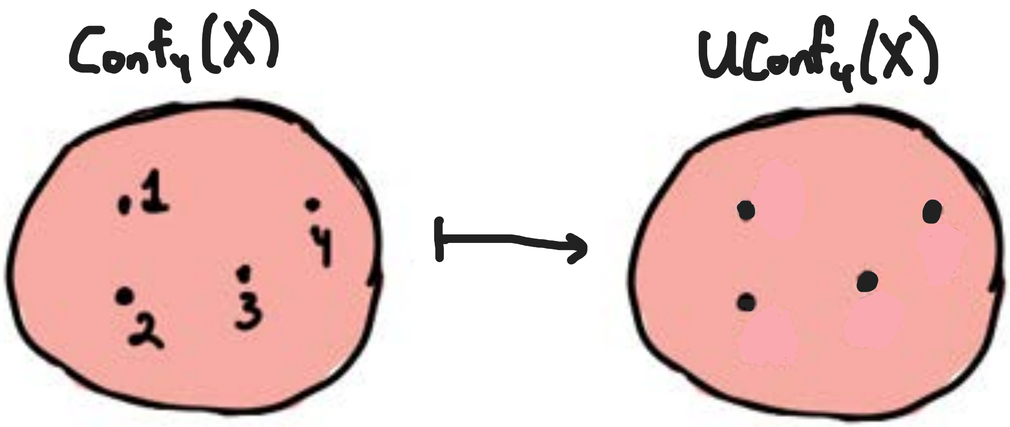


represents a single pt in config. space

We can think of this space as the collection of n distinct & labeled pts on X.

If we forget the labels, we get the unordered config. space, $UConf_n(X) := Conf_n(X)/\Sigma_n$.

a mfld, b/c quotient of a mfld by a free action



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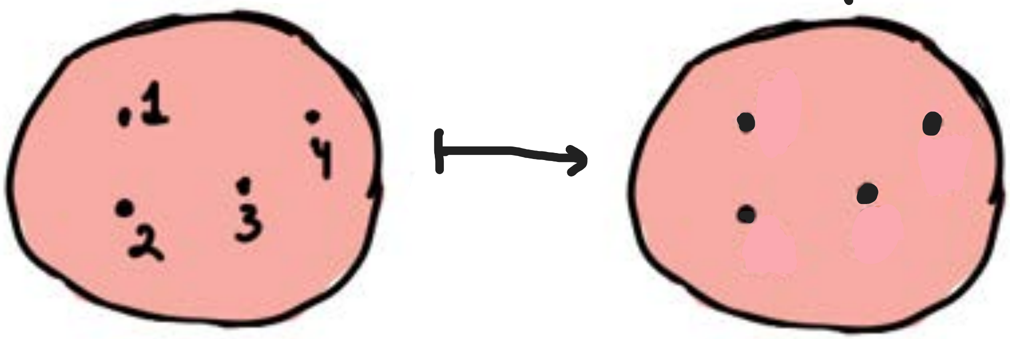
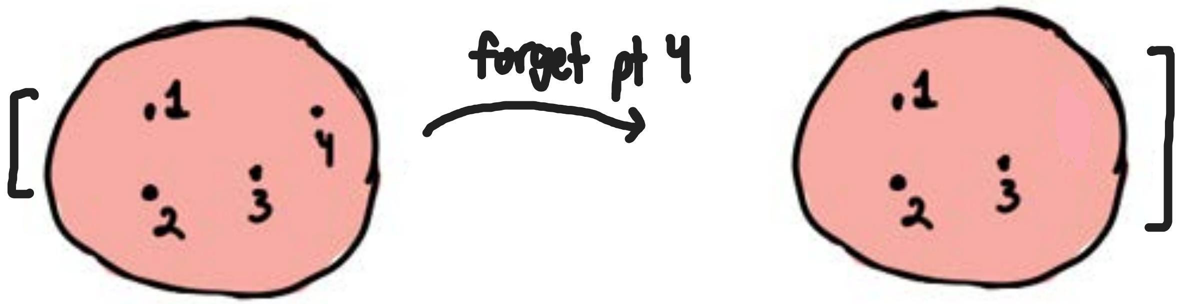
$\mathrm{Conf}_4(X)$ $\qquad\qquad$ $\mathrm{UConf}_4(X)$



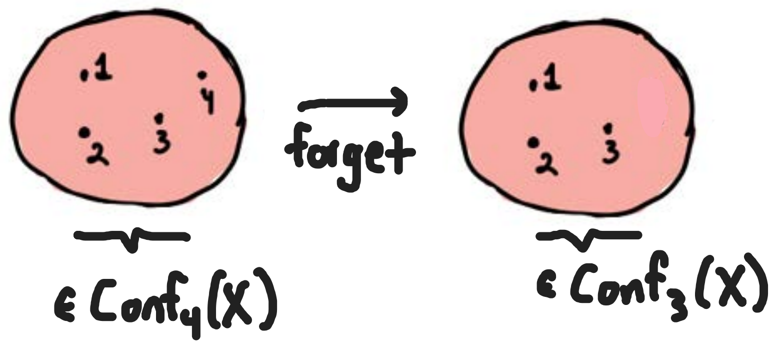
Can also forget a point (in ordered configuration space)

$\in \mathrm{Conf}_4(X)$



$\in \mathrm{Conf}_3(X)$

Let's focus on this second map first...



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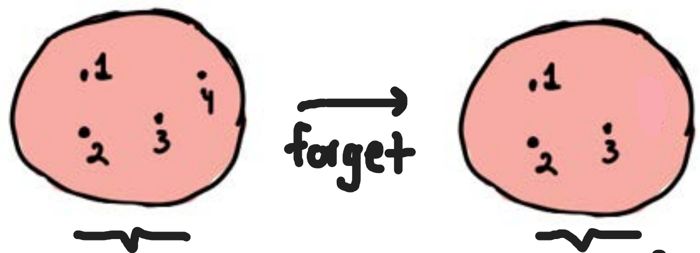

$\in \mathrm{Conf}_4(X) \xrightarrow{\text{forget}} \in \mathrm{Conf}_3(X)$

What's the fiber?

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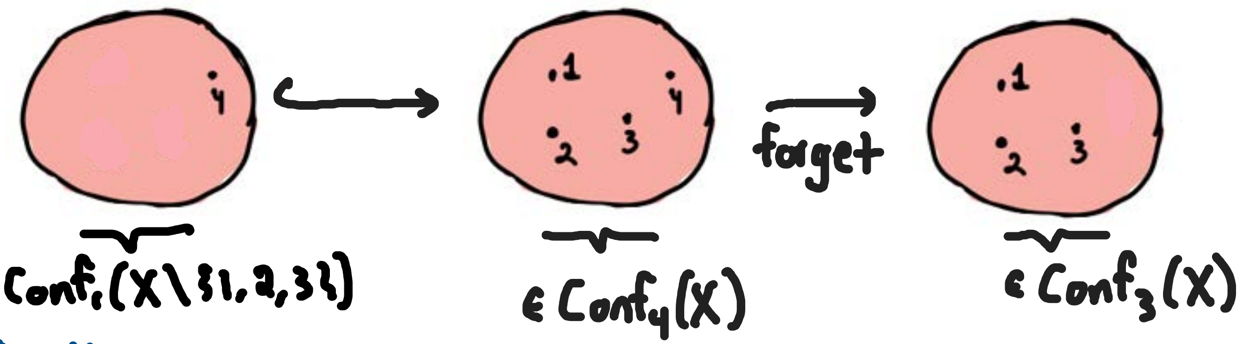

What's the fiber? Fix 3 pts. The 4^{th} can be placed anywhere except for where the first 3 sit.

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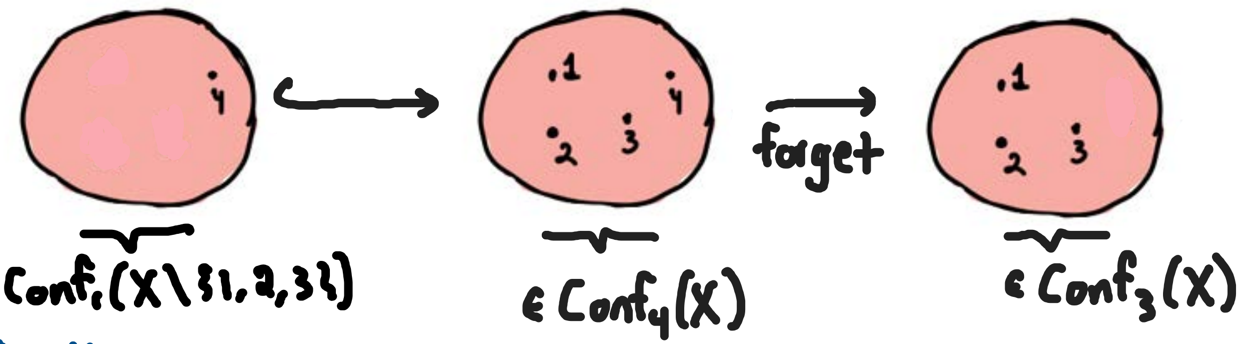

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Turns out: The "forgetting a point" map

$$\begin{aligned} \mathrm{Conf}_{k+1}(M) &\longrightarrow \mathrm{Conf}_k(M) \\ (x_1,\dots,x_{k+1}) &\longmapsto (x_1,\dots,x_k) \end{aligned}$$

is a <u>fiber</u> <u>bundle</u>.

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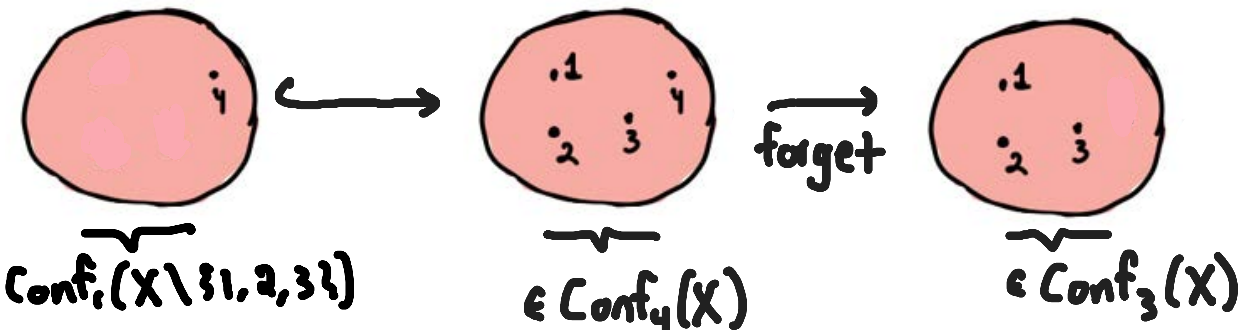

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Turns out: The "forgetting a point" map $\mathrm{Conf}_{k+1}(M) \longrightarrow \mathrm{Conf}_k(M)$ is a fiber bundle.

$$(x_1,\ldots,x_{k+1}) \longmapsto (x_1,\ldots,x_k)$$

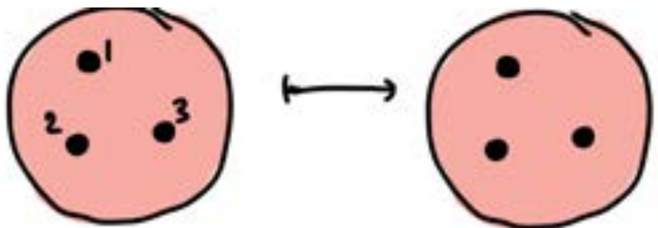
Follows from: $\boxed{\text{Fadell-Neuwirth Thm 1}}$

$\pi: \mathrm{Conf}_n(M) \longrightarrow M$ is a locally trivial fiber bundle, with fiber $\mathrm{Conf}_{n-1}(M\setminus\{y\})$ $(n \geq 2)$.

only remember one point

some pt in M

What about forgetting the ordering?



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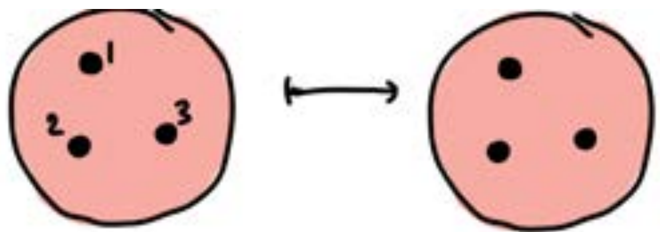

$$\mathrm{Conf}_n(M) \longrightarrow \mathrm{Conf}_n(M)/\Sigma_n = \mathrm{UConf}_n(M)$$

This is also a fiber bundle! (w/ discrete fibers $\cong \Sigma_n$, so a covering space).

(Σ_n acts freely & properly disc. on $\mathrm{Conf}_n(M)$)

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Pf:

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Pf: ① Van-Kampen's Thm + assumption on $n \Rightarrow M \setminus \{\text{finitely many pts}\}$ is s.c. (Express $M = M \setminus \{pts\} \cup (B^n \setminus \{pts\})$, use $B^n \setminus \{pts\} \simeq \bigvee S^{n-1}$ s.c. for $n \geq 3$.)

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② The fibration $M \setminus \{y_1, \dots, y_{k-1}\} \hookrightarrow \mathrm{Conf}_k(M) \longrightarrow \mathrm{Conf}_{k-1}(M)$ yields a LES of htpy gps

$$\cdots \to \pi_1(M \setminus \{y_1, \dots, y_k\}) \longrightarrow \pi_1(\mathrm{Conf}_k(M)) \to \pi_1(\mathrm{Conf}_{k-1}(M)) \to$$

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③ When $k=0$, $\mathrm{Conf}_0(M) = *$. When $k=1$, $\mathrm{Conf}_1(M) = M$, so $k=0,1$ are trivial.

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5. $\pi_1(U\mathrm{Conf}_k(M)) = \Sigma_k$ since $\mathrm{Conf}_k(M) \longrightarrow U\mathrm{Conf}_k(M)$ is a universal cover.

deck gp $= \Sigma_k$

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"Poincaré polygon thm"

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② Assumption on $M \Longrightarrow M \setminus \{y_1, \dots, y_{k-1}\}$ is aspherical (h.e. wedge of circles)

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④ $\mathrm{Conf}_k(M) \to \mathrm{UConf}_k(M)$ covering space $\Rightarrow \pi_i(\mathrm{UConf}_k(M)) = 0 \quad \forall i \geq 2.$

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for M = connected surface other than S^2 or $\mathbb{RP}^2$

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Q: What is π?

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Artin's braid group

Specializing to $M = \mathbb{R}^2$ (or $\mathbb{C}$). How to compute $B_n(\mathbb{R}^2) = \pi_1(U\mathrm{Conf}_n(\mathbb{R}^2))$?

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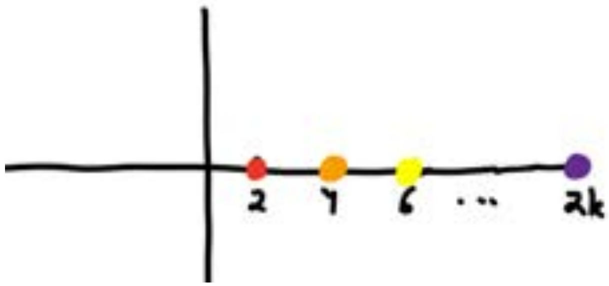
Fix a basepoint $\{(2i, 0)\}_{i=1}^{k}$. $(*)$



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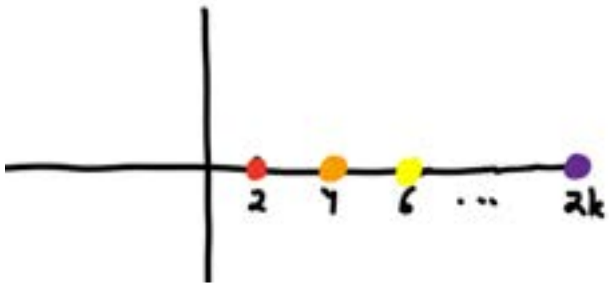

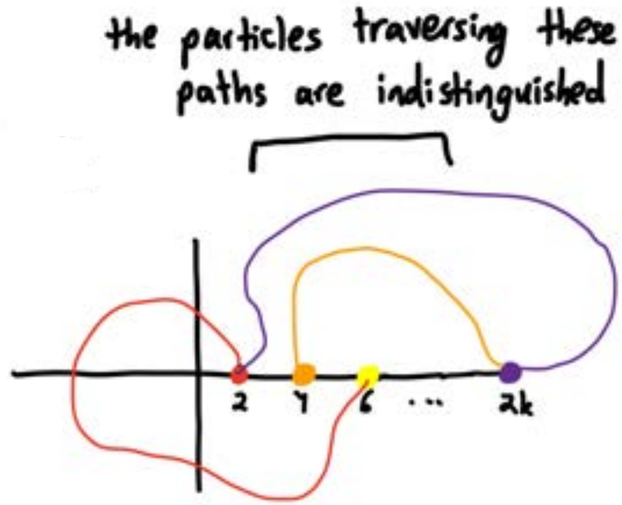

An elt of $B_k(\mathbb{R}^2)$ is a (htpy class of a) path through $UConf_k(\mathbb{R}^2)$ starting & ending at $(*)$.

Can think of this as k particles moving through $\mathbb{R}^2$ & never crashing into each other.

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Fix a basepoint $\{(a_i,0)\}_{i=1}^{k}$. (*)

This means: An elt of $B_k(\mathbb{R}^2)$ is determined by the data of:

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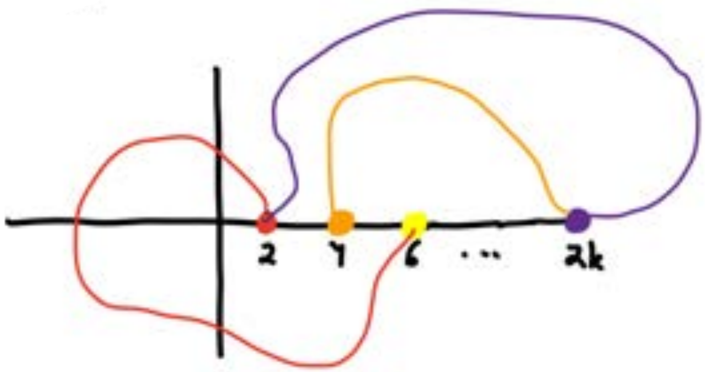

(i) a permutation $\tau \in \Sigma_k$

(ii) a path $p : [0,\pi] \to \underbrace{(\mathbb{R}^2)^k}$

(so really k paths)

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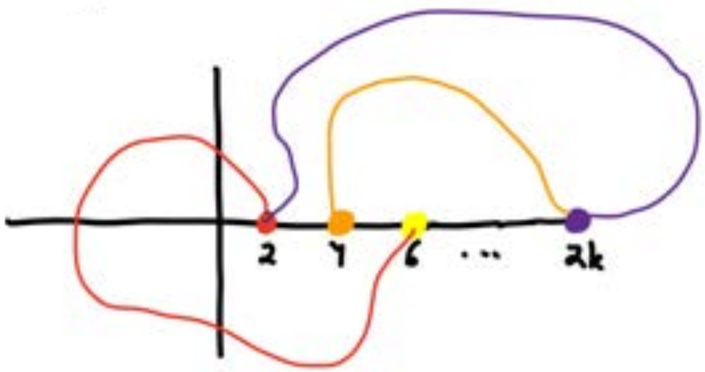

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Satisfying

(1) $(p(0))_r = (2r, 0) \quad \forall \ 1 \le r \le k$

(2) $(p(1))_r = (2 \cdot \tau(r), 0) \quad \forall \ 1 \le r \le k$

(3) $(p(t))_r \ne (p(t))_{r'} \quad \forall \ 1 \le r \ne r' \le k$ & $t \in [0,\pi]$.

Obtain a canonical group homomorphism $\pi_1(UConf_k(\mathbb{R}^2)) \xrightarrow{\pi} \Sigma_k$

Artin's braid group Obtain a canonical group homomorphism $\pi_1(UConf_k(\mathbb{R}^2)) \xrightarrow{\pi} \Sigma_k$

Q: What's the kernel of π?

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Claim: $B_k(\mathbb{R}^2) \overset{\pi}{\twoheadrightarrow} \Sigma_k$. ($\Rightarrow$ SES $1 \to PB_k \to B_k \to \Sigma_k \to 1$)

Pf: Since π is a gp hom., suffices to show generators are attained.

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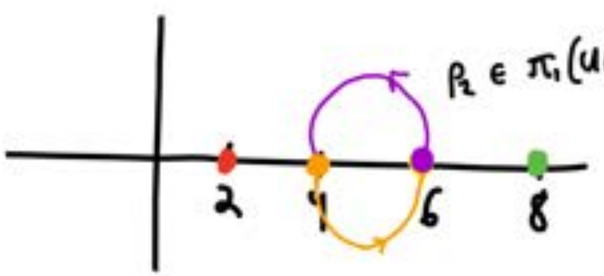
Pf: Since π is a gp hom., suffices to show generators are attained.

For $1 \le i \le r$, define a path $p_i : [0,\pi] \to (\mathbb{R}^2)^k$ via

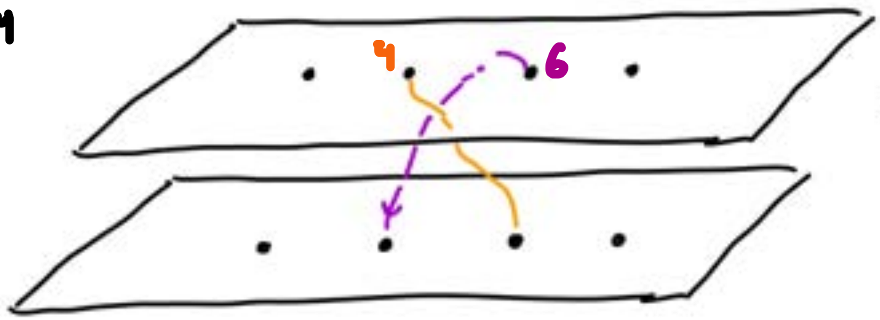
$$(p_i(t))_r = \begin{cases} (2r, 0) & r \notin \{i, i+1\} \\ (2i+1+\cos(t+\pi), \sin(t+\pi)) & r = i \\ (2i+1+\cos(t), \sin(t)) & r = i+1. \end{cases}$$



$\in \Sigma_4$

$\pi(\rho_2) = (2\ 3)$

$\in B_4(\mathbb{R}^2)$



stack $\mathbb{R}^2$'s
imagine strands
moving through

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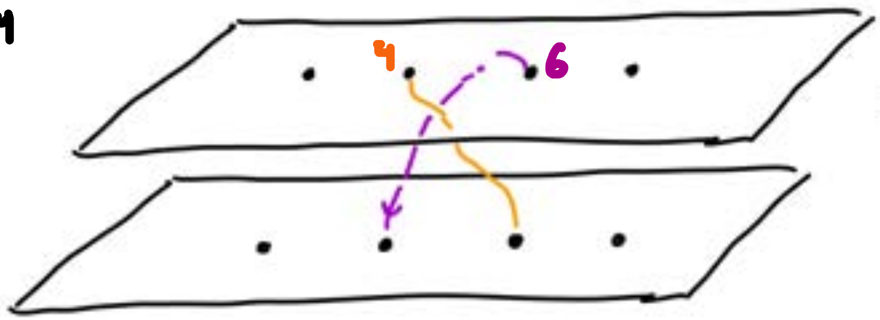

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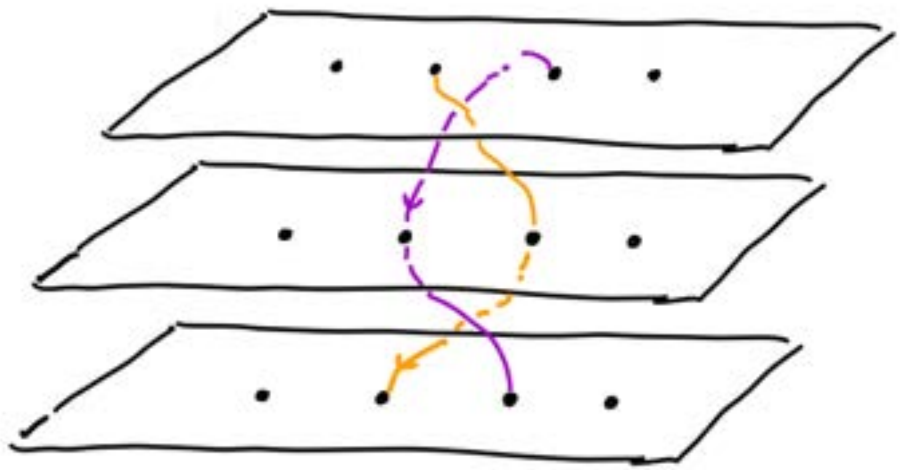

$\in PB_n$

visual demonstration
that loops in $Conf_n(\mathbb{R}^2)$ &
$UConf_n(\mathbb{R}^2)$ correspond to braids

Turns out: if $|i-j|>1$, then ρ_i & ρ_j commute.

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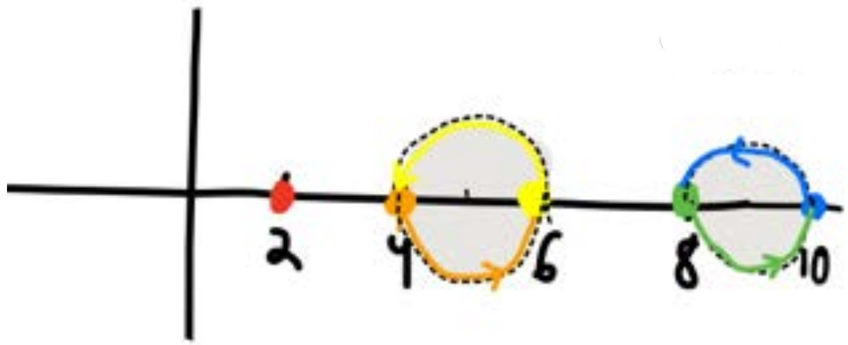
WLOG $j>i$.



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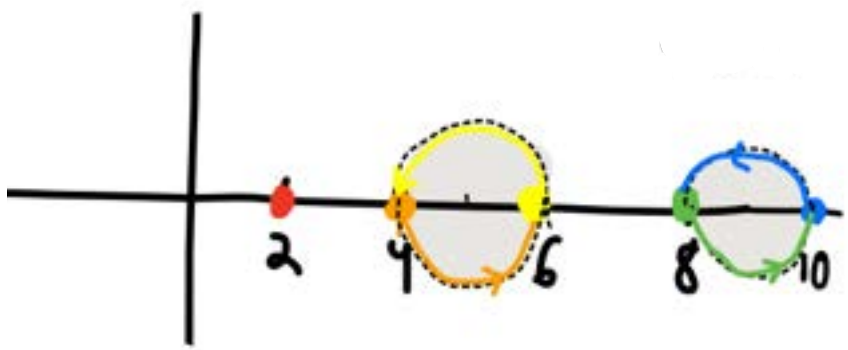
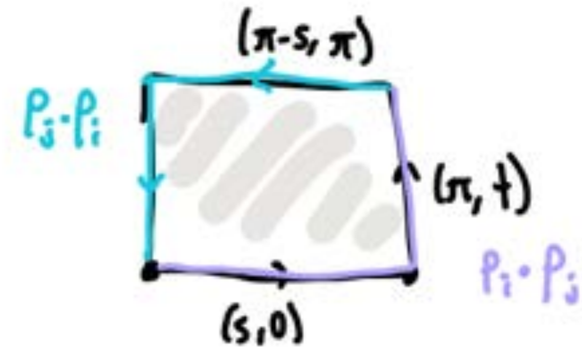
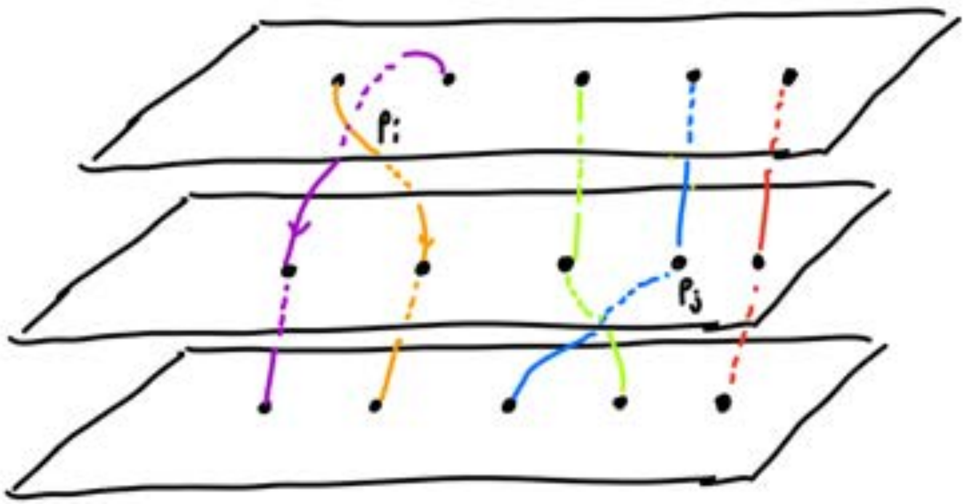


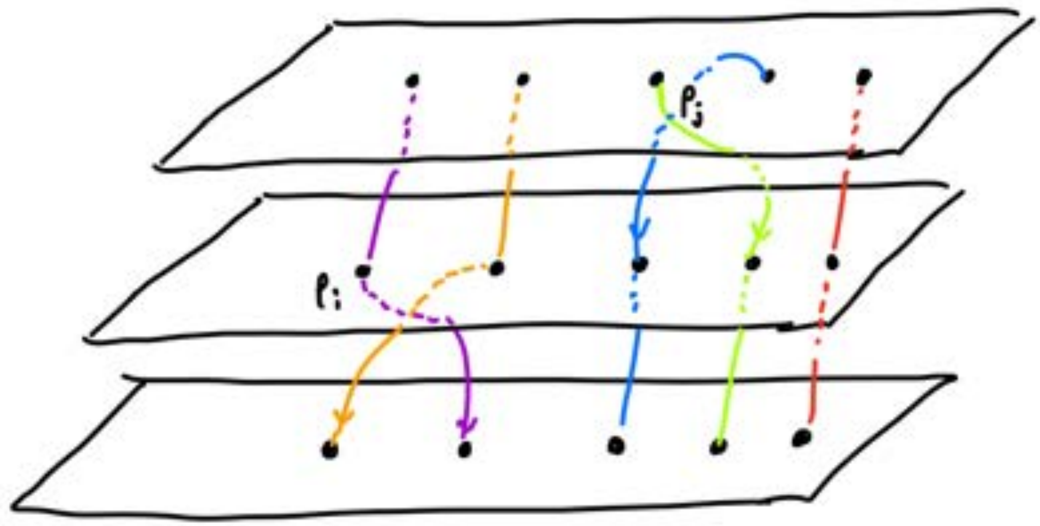
Define

$$H(s,t)_r = \begin{cases} (2r,0) & r \notin \{i, i+1, j, j+1\} \\ (\rho_i(s))_r & r \in \{i, i+1\} \\ (\rho_j(t))_r & r \in \{j, j+1\} \end{cases}$$





p_i
p_j



p_j
p_i

What about braiding adjacent strands?

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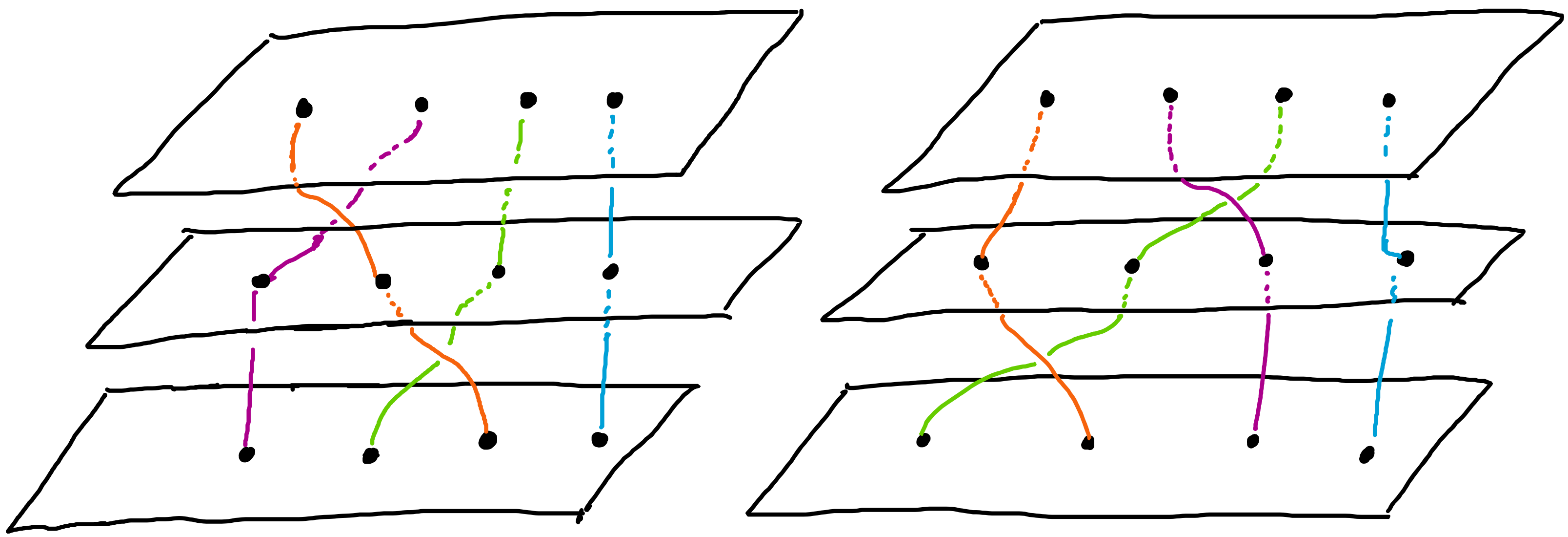
Is $\rho_i \rho_{i+1} = \rho_{i+1} \rho_i$?



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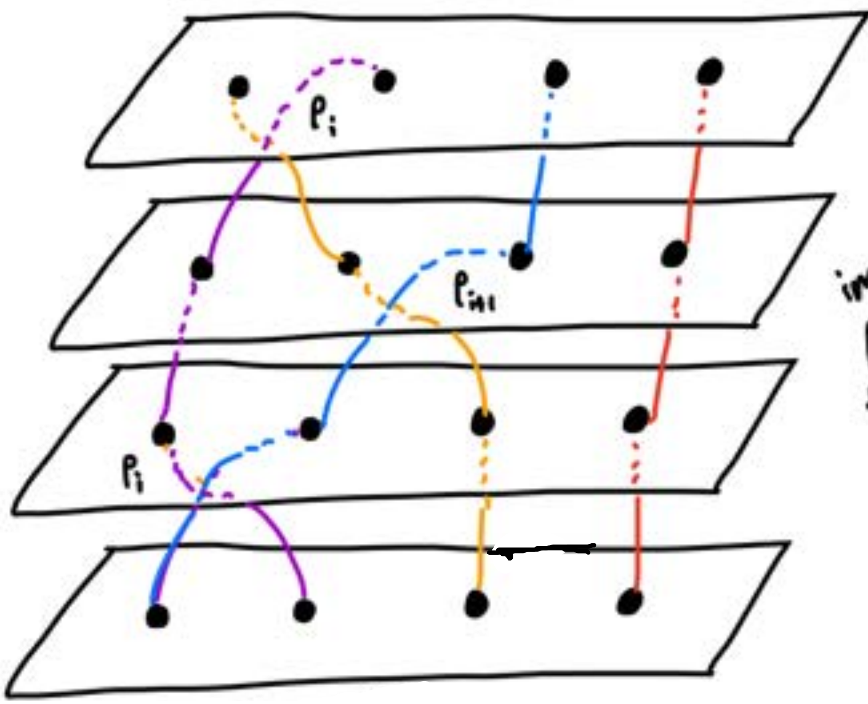
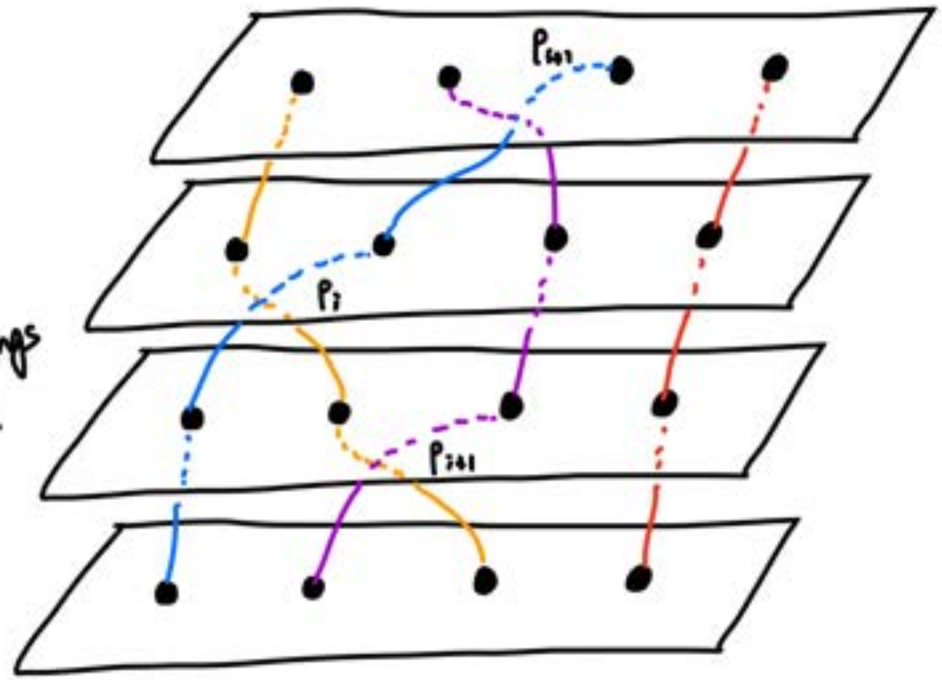


imagine pulling the strings taut!



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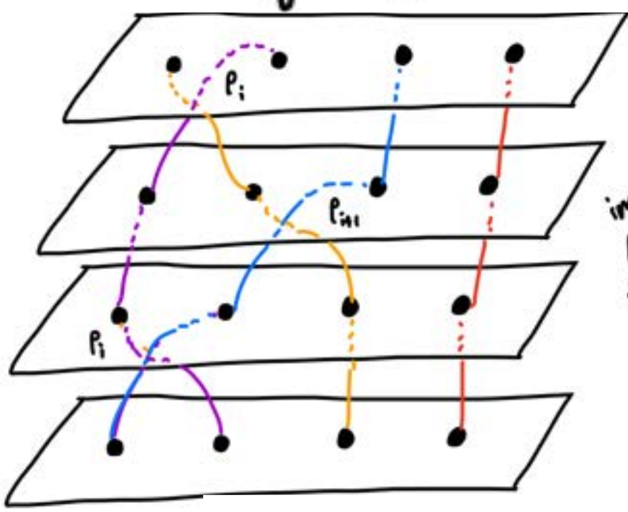

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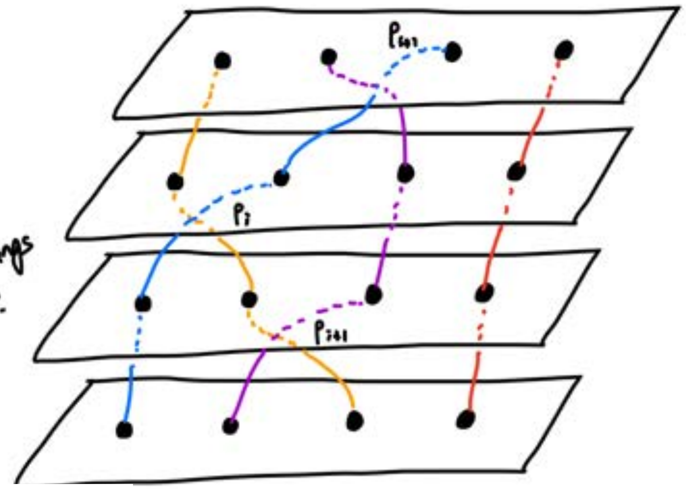
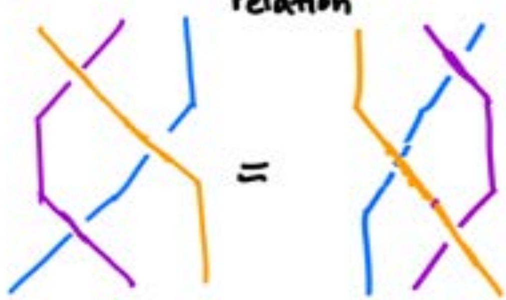


braid relation



$$\rho_i \rho_{i+1} \rho_i = \rho_{i+1} \rho_i \rho_{i+1} \quad \forall\, 1 \le i \le k.$$

\ goes over / goes under

group of "isotopy classes" of braids

the only relations!

Artin's braid group! '47 $B_n = \langle \rho_1, \ldots, \rho_{n-1} \mid \underbrace{\rho_i \rho_{i+1} \rho_i = \rho_{i+1} \rho_i \rho_{i+1}, \quad \rho_i \rho_j = \rho_j \rho_i \text{ for } |i-j| > 1}\rangle$

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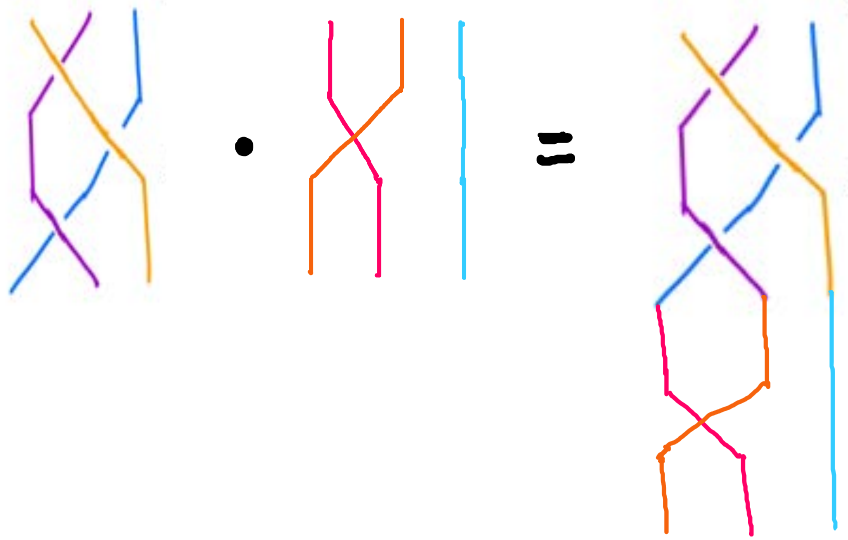
Group operation $\longleftrightarrow$ concatenating braid diagrams



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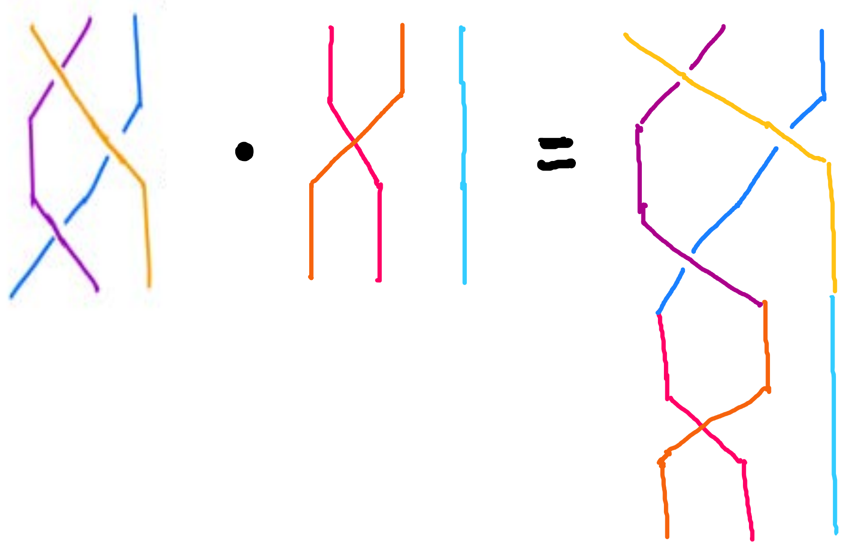
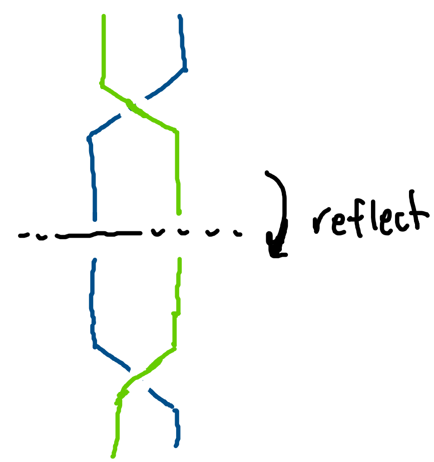


inverses $\longleftrightarrow$ reflect across $\mathbb{R}^2 \times \{0\}$



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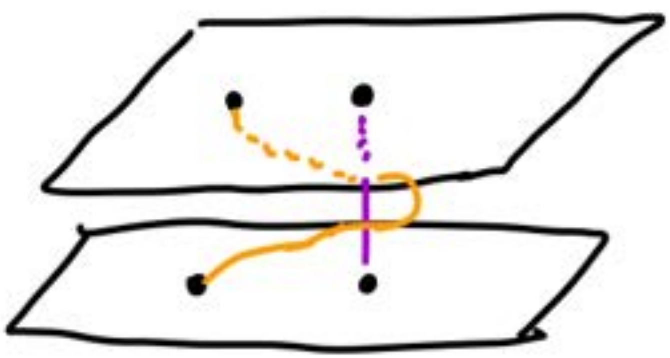
Can do a similar thing for pure braid groups.

Generators $= \{T_{ij}\}$ where string i wraps around string j counterclockwise.



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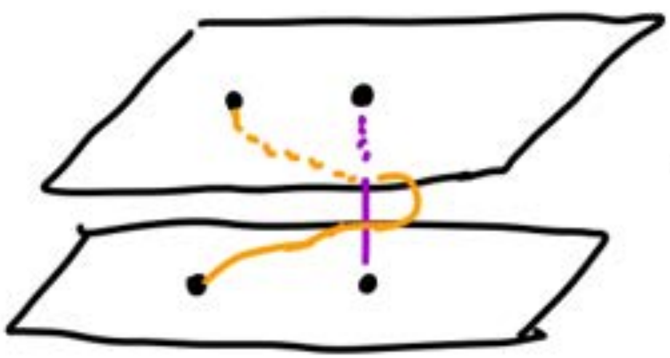

There are finitely many relations and they are all central!

mapping class groups

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Def Let M be a (smooth) oriented manifold. Let $\mathrm{Diff}^+(M, \partial M) = \{$orientation-preserving diffeos $M \to M$ fixing ∂M ptwise$\}$.
Topologize $\mathrm{Diff}^+(M)$ w/ the compact-open topology. Then path-components in $\mathrm{Diff}^+(M) \longleftrightarrow$ isotopy classes of diffeos

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Define $\mathrm{Mod}(M) = \mathrm{MCG}(M) = \mathrm{Diff}^+(M)/\sim$ $\quad$ $f \sim g$ iff $\exists$ isotopy H b/w f & g (H_t diffeo $\forall t$ & fixes ∂M)

$$= \pi_0(\mathrm{Diff}^+(M))$$

mapping class groups

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Alexander Lemma

Fact $\mathrm{MCG}(D^2) = \{e\}$ (triv.)

(if $f: D^2 \to D^2$ is a homeo fixing ∂D^2 ptwise, then $f_t(x) = \begin{cases} (1-t)\cdot f\left(\frac{x}{1-t}\right) & 0 \le |x| < 1-t \\ x & 1-t \le |x| \le 1 \end{cases}$
gives an isotopy of f to the identity map s.t. ∂D^2 is fixed $\forall t$.)

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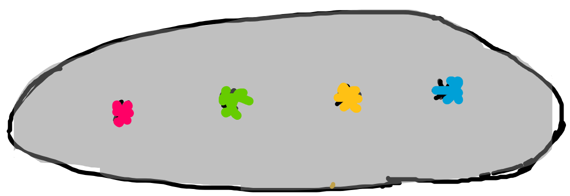
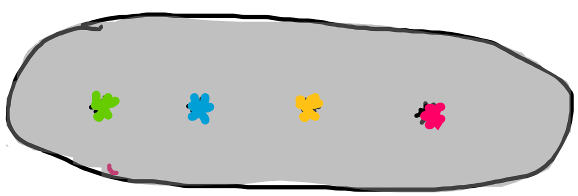


$\swarrow$ $\mathrm{id}(D_n)$



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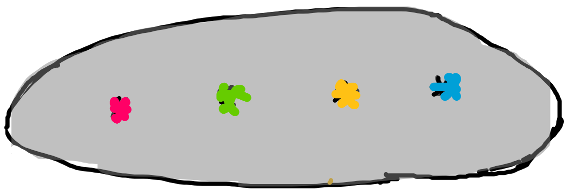

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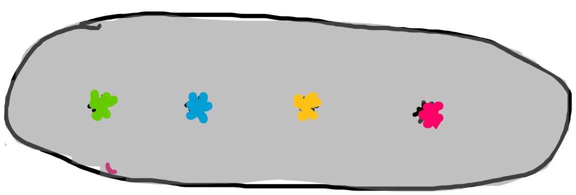

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! Since $\overline{\Phi}$ is a homeo of D_n, Alexander Lemma $\Rightarrow \overline{\Phi} \simeq \mathrm{id}$.

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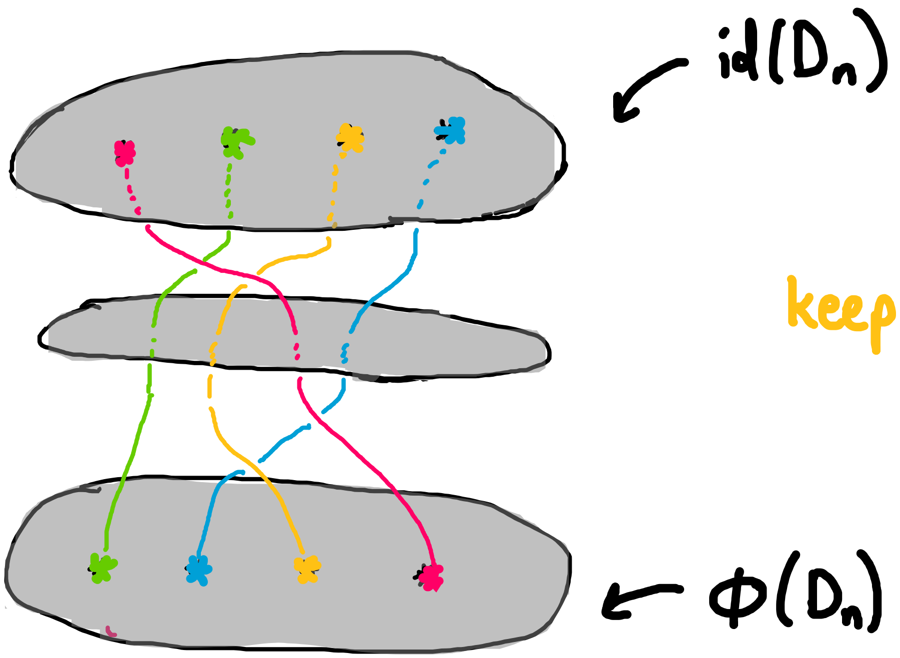

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keep track of where the punctures (marked points) move throughout the isotopy

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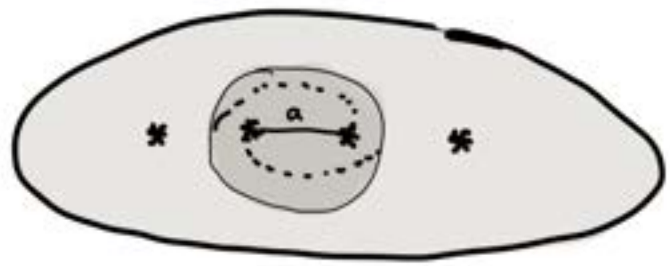
$\rho_i \longmapsto$ half twist H_α



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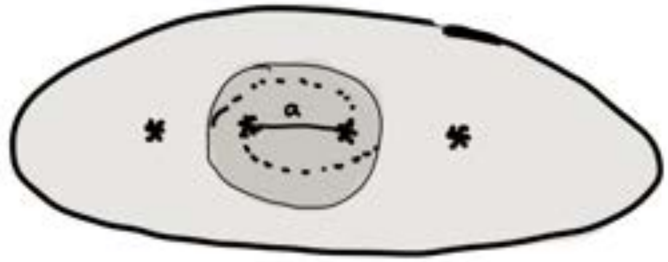
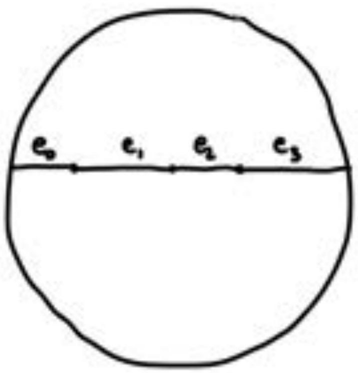
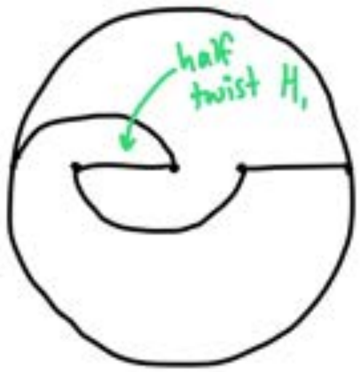


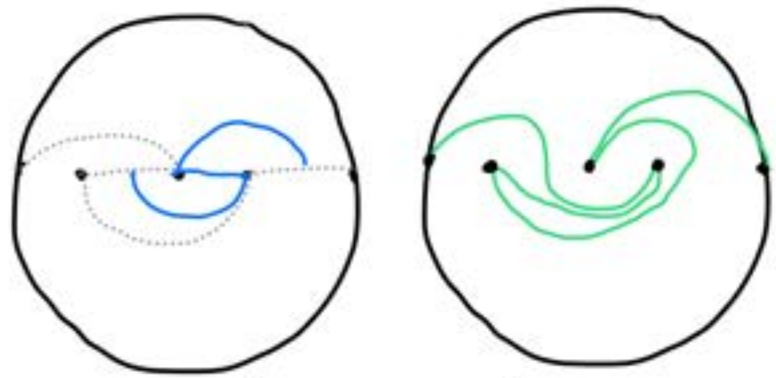
Elements of $\mathrm{Mod}(D_n)$ can be represented by curve diagrams: (taken from *Ordering Braids* by Dehornoy et al)



curve diagram of the trivial braid



curve diagram of σ_1 (braid where strands 1&2 cross)



curve diagram of $\sigma_2^{-1} \cdot \sigma_1$ (apply half-twist σ_2^{-1} to curve diagram to the left)

Integral Burau Representation

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Assign to each configuration $(z_1, \dots, z_n) \in UConf_n(D^2)$ a "hyperelliptic curve"

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has 2 solutions except @ z_i's

$$\parallel$$

$$X_{(z_1, \dots, z_n)}$$

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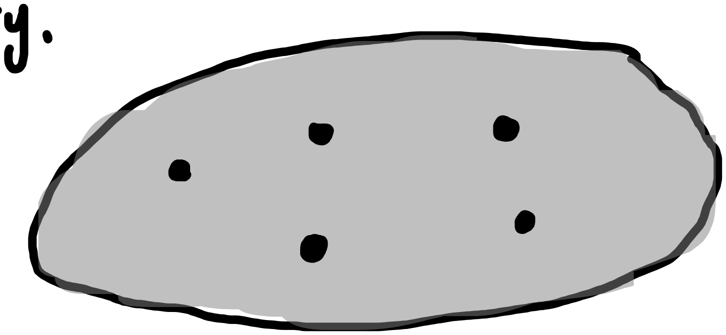
The recipe: Draw the configuration in the disk. Make slits between successive pairs of pts. If an odd #, the last pt is connected by a slit to the boundary.



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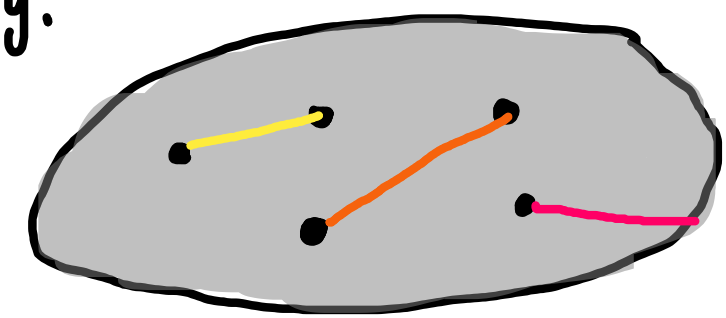
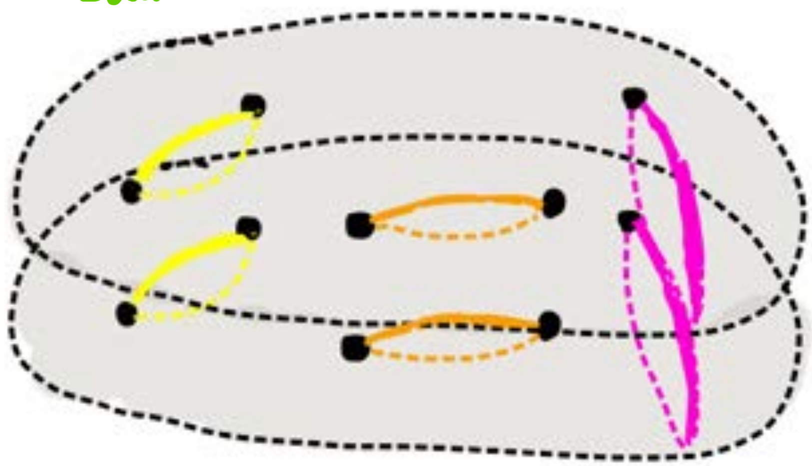
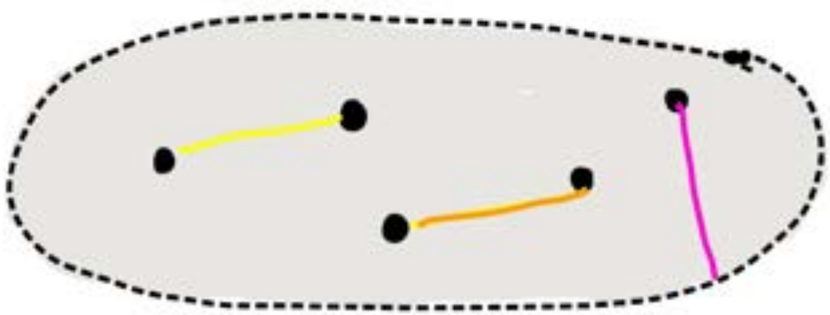


here $n=2g+1$, $g=2$

Make 2 copies of your disk. Glue the sheets so that top slit of top sheet glues to bottom slit of bottom sheet, & vice versa:

backwards





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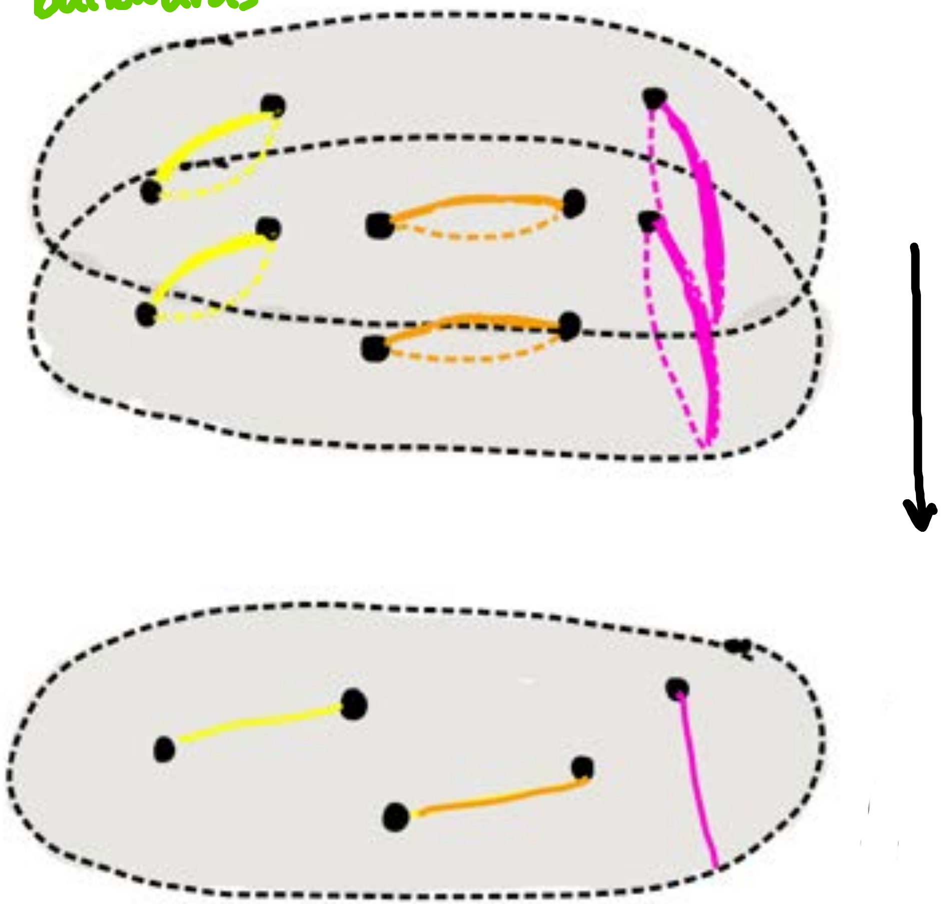
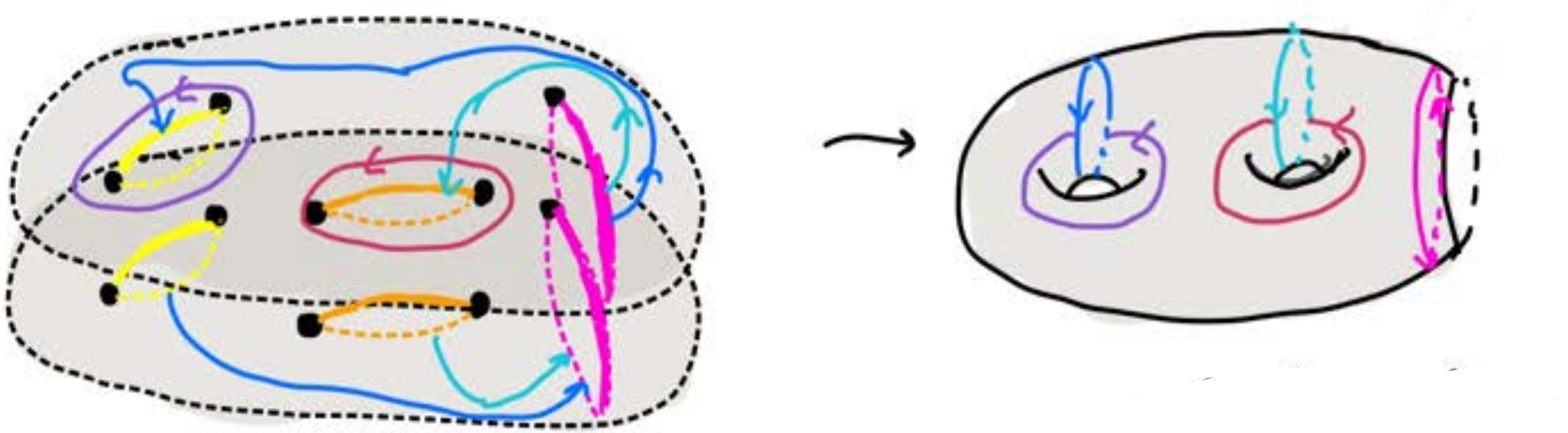


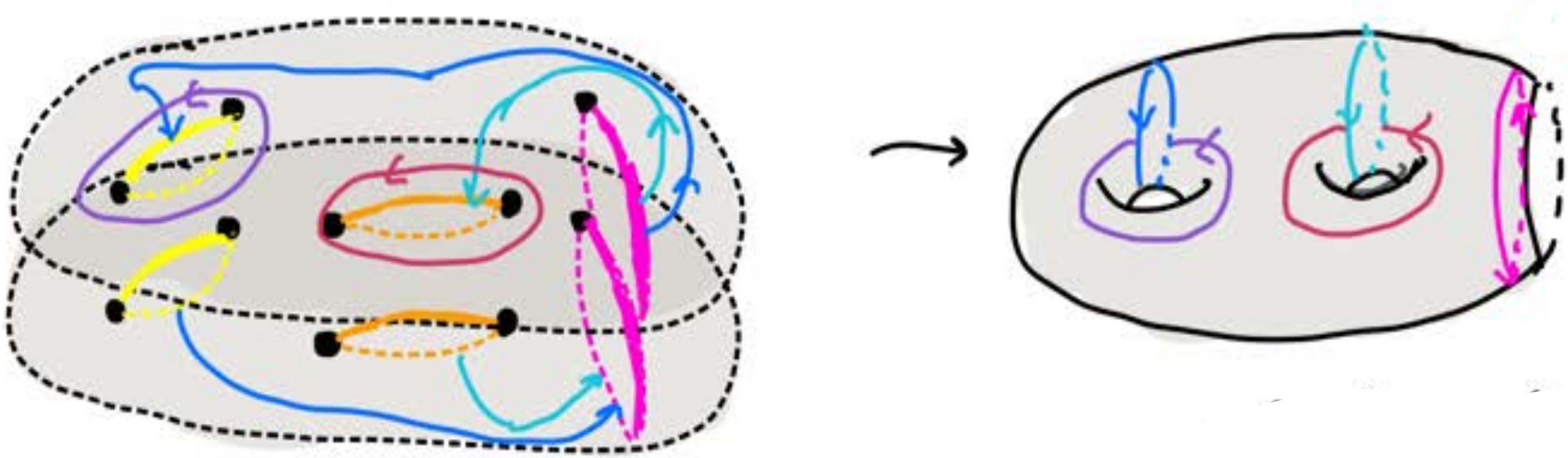
<u>Claim:</u> this gluing yields a genus g surface with 1 boundary component.

How to see? Take a walk around!



"go down one hole, walk across the bottom to the right-most slit, go up that hole, walk across the top back where you started. that produces 1 genus!"

! What is the genus & # of boundary components? $\chi = 2-2g-(b+p)$



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$\boxed{\text{Riemann-Hurwitz}}$

$\chi(\text{upstairs}) = 2(\chi(D^2))$

$- \sum e_p - 1$

$e_p = 2$ for the $2g+\varepsilon$ branch pts

$\varepsilon \in \{1,2\}$

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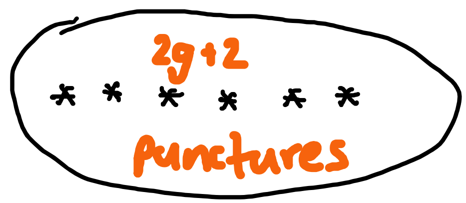

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$2-2g'-b = 2(1)-(2g+1)$
$\Rightarrow -2g'-b = -2g+1$
$\Rightarrow b=1,\ g'=g$

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$\chi(\text{upstairs}) = 2(\chi(D^2)) - \sum (e_p - 1)$

$e_p = 2$ for the $2g+\varepsilon$ branch pts
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$2-2g'-b = 2-(2g+2)$
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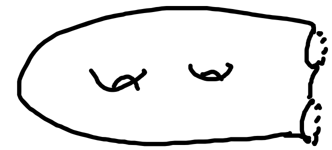

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(get 1 or 2 boundary components, depending on if boundary is a "branch pt") get a fibration $\Sigma_{g,1} \hookrightarrow \underbrace{E}_{} \longrightarrow UConf_{2g+1}(D^2)$

total space, $\{(x, y, \vec{a}) : \vec{a} \in UConf_{2g+1}(D^2),\ y^2 = \prod(x - a_i)\}$

get a monodromy action of $\pi_1(UConf_{2g+1}(D^2)) = B_{2g+1}$ on $H_1(\Sigma_{g,1}) \cong \mathbb{Z}^{2g}$.

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This action preserves the symplectic form on $H_1(\Sigma_{g,1})$ (intersection form).

So $Aut(H_1(\Sigma_{g,1})) = Sp_{2g}(\mathbb{Z})$.

Get a gp hom $B_{2g+1} \longrightarrow Sp_{2g}(\mathbb{Z})$!!!!!!

"integral Burau rep."

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(rarely surjective — image is the integral Burau rep)

THANK

YOU!

happy
birthday
Jeremy!